LocalBook System Architecture — Test Document

1. Overview

LocalBook is a privacy-first AI research notebook built with Tauri, FastAPI, and Ollama.

All processing happens locally on the user's machine using Apple Silicon GPUs.

No data is sent to external servers — the entire AI pipeline runs on-device.

The system architecture consists of five layers:

• Presentation Layer: Tauri webview with real-time UI

• API Layer: FastAPI backend with REST endpoints and WebSocket connections

• Intelligence Layer: RAG engine, context builder, intent classifier, and content generators

• Storage Layer: LanceDB for vector embeddings, JSON files for metadata

• Model Layer: Ollama-managed LLMs including reasoning, embedding, and vision models

2. RAG Pipeline

The Retrieval-Augmented Generation pipeline is the core of LocalBook's AI capabilities.

When a user asks a question, the system:

1. Embeds the query using the Snowflake Arctic Embed 2 model (1024 dimensions)

2. Searches the LanceDB vector store for the top-k most relevant chunks

3. Reranks results using cross-encoder scoring

4. Builds an adaptive context window based on available token budget

5. Generates an answer using the main reasoning model (Gemma 4 E4B by default)

6. Extracts and formats citations from the source material

The embedding model produces 1024-dimensional dense vectors. Each document is chunked into passages of approximately 512 tokens with 50-token overlap. The chunking strategy uses semantic boundaries (paragraphs, headers) rather than fixed character counts.

3. Model Roles

LocalBook uses a multi-model architecture with specialized roles:

Main Model (gemma4:e4b): Handles complex reasoning, document generation, native image understanding, and streaming chat responses. Runs with a context window of up to 16384 tokens. Typical throughput: 15-25 tokens/second on M4 16GB. Apache-2.0 license.

Fast Model (phi4-mini): Used for quick tasks like intent classification, follow-up questions, and simple lookups. 3.8B parameters with MIT license.

Embedding Model (snowflake-arctic-embed2): Generates vector embeddings for semantic search. 568M parameters, produces 1024-dimensional vectors. Apache-2.0 license.

Vision: Gemma 4 E4B is natively multimodal, so it also analyzes images, charts, and diagrams during document ingestion — no separate vision model is required by default.

4. Performance Benchmarks

On an Apple M4 chip with 16GB RAM:

- Gemma 4 E4B: 15-25 tokens/sec average, ~1s time-to-first-token
- Phi-4 Mini: 35 tokens/sec average, 0.6s time-to-first-token
- Snowflake Arctic Embed 2: 120 embeddings/sec for 512-token passages
- Gemma 4 vision: 3-5 seconds per image description

Figure 1: Model Throughput Comparison (tokens/sec)